

Project Design Phase
Solution Architecture

Date	1 July 2026
Team ID	
Project Name	BookStore
Maximum Marks	4 Marks

Solution Architecture:

The Readora Online Bookstore provides a centralized platform for users to browse, search, purchase, and manage books through a secure and user-friendly web application. Built using the MERN Stack (MongoDB, Express.js, React.js, and Node.js), the system connects customers, administrators, and the database through RESTful APIs, ensuring efficient communication and seamless data management.

The primary goals of the solution are to:

- Provide a simple and user-friendly platform for browsing and purchasing books online.
- Enable users to search, filter, and explore books by category, author, or title.
- Allow customers to securely register, log in, manage their cart, and place orders.
- Provide an admin dashboard to add, update, delete, and manage books and customer orders.
- Ensure secure authentication and authorization using JWT and encrypted passwords.
- Store and manage book, user, and order data efficiently using MongoDB Atlas.
- Support scalability and future enhancements such as online payment integration, book recommendations, reviews, and cloud deployment.

Readora- Bookstore – Solution Architecture Diagram:

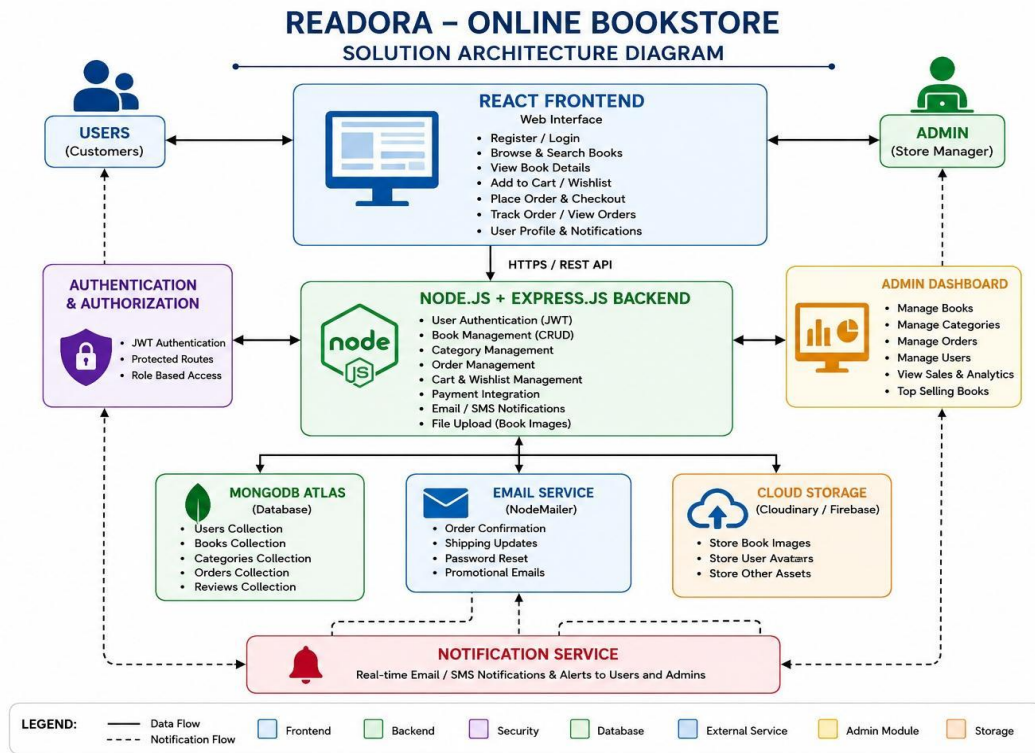


Figure 1: Architecture and data flow of the Readora-Bookstore.

Reference: <https://www.mongodb.com/resources/languages/mern-stack/>